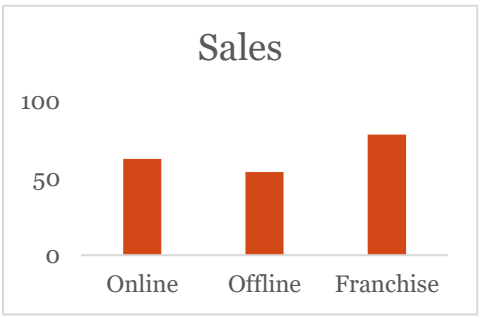
ANOVA (Analysis of Variance)

Definition:
ANOVA(Analysis of Variance) is statisctical test used to determine whether the population means of three or more independent groups are equal.

Question: Which store type generates different averag monthly sales?

Store	Sales
Online	62.6
Offline	54.2
Franchise	78.4

The averages look different. ANOVA tests whether the observed differences are statistically significant rather than merely due to sample variation.



H0:μ1=μ2=μ3
H1: At least one population mean differs

F- Statistic and the Core Logic

The F-Statistic compares variation between groups with variation within group.

$$F=\frac{MS_{Between}}{MS_{Within}}$$

1. Between group variation: How far the group means are from the overall mean
2. Within-group variation: How much individual observations vary inside each group.

Betweeen Groups:
Are group averages far apart?
Within groups:
How much random variation exists inside each group?

$$F=\frac{\text{Between-group variation}}{\text{Within group variation}}$$

A larger F-Statistic indicates stronger evidence against equal populaiton means, assuming the ANOVA conditions hold.

Question If the group means are very different but the observations inside each group are tightly clustured, what happens to F?
Answer: F tends to increase

Significance Level

$$\alpha=0.05$$

This is the chosen threshold for the probability of a Type 1 error under the null hypothesis. It is not the probability that the null hypothesis is true.

One Way ANOVA:

Business Problem:

A retail company compares monthly sales across three store types. Each group contains five stores The sales values are in ₹ thousands.

Store	Online	Offline	Franchise
1	60	50	80
2	65	55	75
3	58	52	77
4	68	60	82
5	62	54	78
Total	313	271	392

Step 1 : Calculate each group mean

$Mean_{Online}$	=313/5	62.6
$Mean_{Offline}$	=271/5	54.2
$Mean_{Franchise}$	=392/5	78.4

Step 2: Calcualate the overall mean

$Mean_{Overall}$	=(313+271+392)/15	65.066667 ≈65.07

Step 3: Calculate between-group sum of squares (SSB)

SSB measures variation due to differences between group means.

$$SSB = \sum n_i(x_i bar - xbar)^2$$

$$\sum n_i \left(\bar{x}_i - \bar{x}\right)^2$$

For Online:

$5(62.6-65.07)^2$ 30.5045

For Offline:

$5(54.2-65.07)^2$ 590.7845

For Franchise

$5(78.4-65.07)^2$ 888.4445

Therefore,

SSB = 1509.734

Step 4: Calculate within-group sum of squares(SSW)

SSW measures variation of obseravtions around their own group mean.

For Online:

SSW_{Online}	$\sum(xi - 62.6)^2$		63.2
		6.76	
		5.76	
		21.16	
		29.16	
		0.36	
$SSW_{Offline}$	$\sum(xi - 54.2)^2$		56.8
		17.64	
		0.64	
		4.84	
		33.64	
		0.04	
$SSW_{Frenchaise}$	$\sum(xi - 78.4)^2$	29.2	
		2.56	
		11.56	
		1.96	
		12.96	
		0.16	

Add these

SSW = 63.2+56.8+29.2 149.2

Step 5: Degreees of freedom

Let k = 3 groups and N = 15 observations

Between groups:

df_B =k-1 =3-1 =2

Within Groups:

df_W =N-k =15-3=12

Total:

df_T =N-1 =15-1 =14

Step 6: Calculate the Mean Squares

MS_B =SSB/df_B	=1509.734/2	754.867
MS_W =SSW/df_W	=149.2/12	12.43333

Step 7: Calculate F

F =754.867/12.4333 60.713164

Step 8: Calculate Fcritical

alpha = 0.05

df_1 = 2(between groups) Numerator Degree of freedom

df_2 = 12(within groups) Denominator degree of freedom

$F_{critical}$ = 3.885

Note:	In hypothesis testing, if the critical value is greater than the calculated value then, we are failed to reject the null hypothesis If the Calculated value is greater than the critical value then we will reject the null hypothesis

$F_{critical}$ = 3.885 $F_{calculated}$ = 60.713

$F_{calculated} > F_{critical} \Rightarrow$ Reject H0 60.713>3.885

Decision: Reject H0. There is statistically significant evidence that not all three population mean sales are equal.

3.885294 0.00000053

Business Interpretation

At least one store type has a different population mean sales level. The overall ANOVA result does not establish that every store types differs from every other one.

Assumptions of ANOVA

1. Normality
2. Independence
3. Equal variances

<u>Post Hoc Tests Tukey and Bonferroni</u>

Why do we need post hoc tests?

ANOVA tells us whether there is evidence that at least one group mean differs. It does not identify the specific groups.

Example:

Online vs offline

Online vs frenchise

offline vs frenchise

Tukey HSD

Tukey's Honestly Significant Difference procedure compares all pairs while controlling the family wise error rate.

For equal sample sizes:

HSD = qcritical*SQRT(MSw/n)

Using MSw = 12.433, n = 5, and q critical = 3.77 for three groups k = 3 , 12 error degree of freedom

=3.77*SQRT(12.433/5)
5.944896731
HSD = 6

All three pairs differ significantly under the Tukey comparison.

Online vs offline	63.2-56.8	6.4
Offline vs Frenchise	56.8-29.2	27.6
Online Vs Frenchise	63.2-29.2	34

Bonferroni Method
Bonferroni controls the overall error rate by adjusting the significance level for multiple comparisons

$\alpha_{adjusted} = \alpha / m$

For three comparisons:

$\alpha_{adjusted} = 0.05/3 = 0.0167$

Tukey: Designed for all pairwise mean comparisons
Bonferroni: flexible adjustment for a set of multiple tests; often more conservative.

Business Applications:
Marketing: Compare sales from three advertisement campaigns
Retail: Compare average sales across different store types
HR: Compare employee satisfaction across the departments.
Operations: Compare average production output across machines
Finance: Compare average returns across investment stretegies
Healthcare: Compare average outcomes across treatment groups.

Why use Tukey's Test?
To identify which pairs of group means differ after a significant overall ANOVA.